P Deepak Reddy

ee23btech11047@iith.ac.in | deepakreddy1510@gmail.com | +91 9032124980 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.15
Sri Chaitanya Junior Kalasala	Class XII (TSBIE)	2023	97.90%
Sri Chaitanya School	Class X (TSSSC)	2021	100.00%

WORK EXPERIENCE

AI Developer Intern | Diagonal Matrix

- Contributed to an AI-driven Data Model Accelerator that converts business requirements into data models, SQL schemas, synthetic datasets, and analytics-ready outputs.
- Implemented automated workflows for **schema generation**, **validation**, synthetic data generation, and **PostgreSQL integration**, including DDL-based validation of PK/FK constraints.
- Developed and tested SQL-based workflows in JAI/Langflow and integrated the generated PostgreSQL database with **SQL Agent workflows** for natural-language data retrieval.

PROJECTS

Data Agent Github

- Developed a router-based multi-agent system using LangGraph and LangChain to classify natural-language requests and route them to specialized **SQL analysis and ETL workflows**.
- Implemented a schema-aware **natural language-to-SQL pipeline** with PostgreSQL schema retrieval, SQL generation, read-only safety validation, query execution, and natural-language result generation.
- Built an ETL workflow that extracts data from APIs, applies **Pandas-based transformations**, and uses LLM-guided tool selection to process data based on user requests.

Document Retrieval System Github

- Developed a multi-document retrieval pipeline using page-aware PDF extraction, token-based chunking, and semantic embeddings stored in PostgreSQL with pgvector.
- Implemented dense vector search with **CrossEncoder reranking** to retrieve relevant passages and generate grounded answers with document and page-level source citations.
- **Evaluated** retrieval performance using **Hit@K, Precision@K, Recall@K, and MRR** to measure and compare the quality of retrieved results.

Credit Card Fraud Detection Github

- Developed a fraud-detection pipeline for highly imbalanced transaction data using structured preprocessing and feature analysis.
- Evaluated **Logistic Regression, Random Forest, and XGBoost** using stratified cross-validation and fraud-focused evaluation metrics.
- Tuned the selected model and optimized the classification threshold on validation data before final test evaluation.

Neural Network Classifier Github

- Implemented a **3-layer feedforward neural network** from scratch using Python and NumPy to classify handwritten digits from the MNIST dataset.
- Evaluated different **activation functions and optimization algorithms**, comparing their impact on model convergence and classification performance.
- **Applied regularization techniques** including weight decay, dropout, and early stopping to improve model generalization and reduce overfitting.

SKILLS

- **Programming Languages:** C++, Python, SQL

- **Core CS**: Data Structures & Algorithms, Operating Systems, Object-Oriented Programming, DBMS
- **AI/ML:** Machine Learning, LLMs, AI Agents, RAG, Embeddings, Semantic Search
- **ML Libraries/Frameworks:** NumPy, Pandas, Matplotlib, scikit-learn, PyTorch, LangChain, LangGraph
- **Databases:** MySQL, PostgreSQL
- **Others:** Git/Github, Docker, Confluence, LaTeX

RELEVANT COURSES

- **Computer Science Courses:** Introduction to Programming, Data Structures and Applications, DBMS
- **Mathematics:** Calculus, Complex Variables, Differential Equations, Vector Calculus, Probability and Random Processes, Matrix Theory, Linear Optimization, Convex Optimization
- **AI/ML:** Introduction to AI/ML, Pattern Recognition and Machine Learning, Statistical Inference, Deep Learning, Predictive Modeling

POSITIONS OF RESPONSIBILITY

Event Coordinator - National Service Scheme , IITH

- Coordinated community outreach initiatives and events involving 2,500+ NSS volunteers, overseeing planning and execution.
- Managed volunteer teams and coordinated activities to ensure effective event execution and engagement.